

Exercise 1. Let X be an infinite dimensional Banach space. Show that a compact map $C : X \rightarrow X$ with closed range must satisfy $\dim CX < \infty$.

Exercise 2. Solve exercise 2 on page 234.

Exercise 3. Solve exercise 8 on page 243.

Exercise 4. Solve exercise 9 on page 243.

Exercise 5. Let $u : \mathbb{R} \mapsto \mathbb{C}$ be a bounded continuous function. Show that the multiplication operator M_u defined on $L^2(\mathbb{R})$ by

$$M_u f = u f$$

cannot be compact on $L^2(\mathbb{R})$ unless if $u = 0$. Moreover, the same holds when $u \in L^\infty(\mathbb{R})$.

Exercise 6. Let H be a separable Hilbert space and let $T : H \rightarrow H$ be a linear map with the property that there exists an orthonormal basis $\{e_n, n \geq 1\}$ of H such that

$$\sum_{n \geq 1} \|T e_n\|^2 < \infty.$$

a) Show that in this case

$$\sum_{n \geq 1} \|T f_n\|^2 < \infty$$

for any other orthonormal basis $\{f_n, n \geq 1\}$ of H , in fact the value of the above sum is independent of the basis. Such maps are called *Hilbert-Schmidt* operators.

b) Show that every Hilbert-Schmidt operator is compact.

Exercise 7. Show that Hilbert-Schmidt operators on $L^2(\mathbb{R})$ coincide with integral operators with a square-integrable kernel, i.e. those of the form

$$f \mapsto \int_{\mathbb{R}} K(x, y) f(y) dy, \quad \text{with} \quad \int_{\mathbb{R}} \int_{\mathbb{R}} |K(x, y)|^2 dy dx < \infty.$$